

Factors influencing energy production in solar panels

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Abstract: Today, production facilities are constantly improving in terms of efficiency, as solar and wind energy, which is identified with the concept of nature-friendly green energy, is abundant and unlimited. Solar energy system technologies have developed greatly, especially in the last 20 years, and production facilities have increased gradually. Wind energy systems have larger energy requirements than solar energy systems. One of its disadvantages is that wind formation cannot be predicted. The position of the sun and the predictability of when and to what degree it will emit light enable production with less margin of error than wind energy systems. The production of energy obtained from factors such as sun and wind varies over time. For a stable energy production, it must be constantly monitored and managed. The communication via power line (PLC) method is preferred in Smart production systems because it uses the existing power line and has low hardware costs. In this study, parameters such as temperature, panel temperature, humidity, light rate, panel current, panel voltage, which are among the factors affecting the efficiency of solar panels, were measured. The change in the temperature behind the panel depending on the outside air temperature, solar radiation and wind speed was investigated. The obtained data was analyzed and the factors affecting energy production were examined.

Keywords: Solar panels, solar energy, Temperature, Affecting factors, Photovoltaic panels.

1. INTRODUCTION

The development of technology and the continuous increase in the world population mean that the need for energy is also increasing. In today's world, the decrease in fossil fuels, which are the source of conventional energy, causes clean and reliable energy to become even more important. Developed countries aim to progress energy policies and increase the share of clean energy in total energy production in order to meet the emerging energy needs. Energy demand per capita in the world is one of the basic indicators of living standards. Energy consumed per person is generally indicative of a higher quality of life. Traditional fuel sources (fossil fuels, hydro, nuclear energy sources, etc.) mostly meet a large part of today's energy demand. However, the known fact is that these resources will be insufficient to meet the energy demand in the future and some of them will be depleted. Its negative effects on the environment are its role in the extinction of many living species. For these reasons, the importance of renewable energy sources is increasing day by day. It is of great importance for our country to disseminate studies on energy production from renewable energy sources.

Renewable energy sources are energy sources such as hydro, solar, wind, geothermal, biomass and tidal type. These are clean, cheap, abundant and environmentally friendly resources. Among these, solar energy is one of the most widely used sources. Solar energy is an endless form of clean energy that can be found on earth and even in space. Directly from solar energy via solar panel electricity is produced and used. Solar panels consist of photovoltaic (PV) cells. PV cells are produced in various types and structural features. The most commonly used types today are polycrystalline and monocrystalline types, and their efficiency generally varies between 15-17%. There are many experimental and theoretical studies on increasing the efficiency of solar panels. The power of solar panels does not only depend on the efficiency of the cells. It also depends on the number of cells in the panels. Depending on the number of cells, a panel is produced in different powers. This also provides significant flexibility in terms of the electrical power needed for a building. A mini solar power plant (mGES) is

established to meet the electricity needs of many places such as residences, public and private workplaces, and industrial establishments with a grid-connected PV system. Especially in developed countries, solar panels consisting of photovoltaic (PV) cells that can directly produce electricity from solar energy are widely used. Approximately 1-2% of the electrical energy produced in the world is produced by PVs [12].

It is known to increase the efficiency of the energy production process and the preparation of solar panels on the basis of graphene-based samples [1-42].

One of the applications used to convert solar energy into electrical energy is the solar panel. These devices, which convert the sunlight falling on them into DC electrical energy, are called solar panels. It also appears as solar cell, photovoltaic panel, solar cell and solar module. The efficiency of solar panels, which are widely used today, varies depending on their structure and this value is between 5% and 20%.

The fact that energy conversion efficiency is not high in solar panel technologies makes it very important to use the electrical energy obtained in the best way.

Since solar panels have non-linear current-voltage characteristics, solar panel voltage and power vary depending on the current drawn. In addition, the amount of energy produced changes as the characteristics of solar panels depending on factors

such as the intensity of the sunlight falling on their surfaces, the panel angle and the temperature of the environment [3].

The Sun is a ball of gas composed of hydrogen and helium..

With the reaction called Nuclear Fusion, which occurs due to the pressure and temperature, hydrogen nuclei are transformed into helium nuclei, and a great deal of radiant energy is released during this transformation

Although the resulting energy is 1360 W/m² outside the world, the amount of energy reaching the earth's surface under the influence of the atmosphere is between 0-1110 W/m². Even a small portion of the energy coming to the earth is much more than the energy used by humanity. Activities related to solar energy gained momentum after 1975. Due to the advancement of technology used in this field and the decrease in costs, it has become available all over the world. There is a distance of 150 million km between the Sun and the Earth. The energy coming to our Earth from the Sun is approximately 20 thousand times the energy consumed on Earth in a year. Not all solar energy can reach the earth's surface; as seen in Figure 1.1, approximately 30% hits the atmospheric layer and reflects back. Approximately 50% of the radiant energy can pass through the atmosphere and reach the earth's surface.

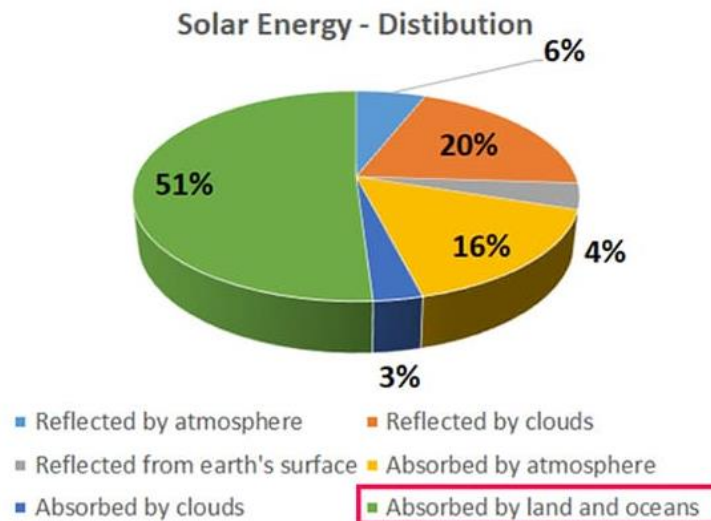


Figure 1. Distribution of Solar Energy [13]

Solar energy technologies vary widely in terms of method, material and technological level and can be divided into two basic groups:

1. Thermal solar technologies: In thermal solar technologies, heat is obtained primarily by utilizing

solar energy. This heat can be used directly or to produce electricity.

2. Solar cells: These semiconductor materials, also called PV batteries, convert sunlight directly into electricity.

The smallest structural units of a solar cell are called cells. When modules come together, they form panels. The panels come together to form arrays. Figure 2 shows the schematic representation of the progression of a solar cell from its smallest structural unit to panel formation.

Dozens of materials are used in making solar cells. Hundreds of materials are subjected to research and

development under laboratory conditions in order to popularize their use. Material selection for photovoltaic cells is of great importance in terms of producing solar cells that are both economical and highly efficient. Silicon is the most commonly used material in solar cell production.

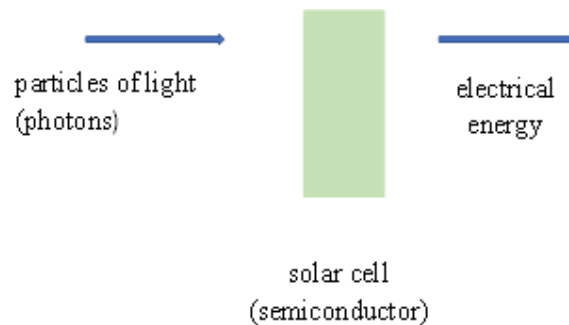


Figure 2. Working principle of solar cells [14]

It can be obtained from mono polycrystalline or sheet and can be produced from sliced thick crystal material or from multiple crystal or thin film layers formed on a carrier [9]. The main materials used in solar cell production are:

- Thick crystalline material: Crystalline silicon, gallium arsenic (GaAs).
- Thin film material: Amorphous silicon, cadmium sulfide (CdS), cadmium telluride (CdTe), copper indium di selenoid (CuInSe2) [9].

The fact that most of today's market activities are directed to the production of monocrystalline silicon semiconductor materials has resulted in technologically advanced monocrystalline semiconductor material production techniques. Monocrystalline Solar Panels were the first commercial solar panels. Single crystal silicon grown by crystal drawing method is used. It constitutes a very large part of the market.

However, the high production cost has increased the tendency towards other batteries. Its efficiency is quite high [1]. It provides 24% efficiency in monocrystalline silicon solar cells under laboratory conditions. In commercial modules, it varies between 15 and 18%. It is ideal for long-term investments due to its high efficiency. The payback period is 4-6 years. 7% efficiency losses occur over a 20-year period [10]. The most commonly used method in the production of polycrystalline silicon is the casting method. The starting material is the same in both monocrystalline

silicon and polycrystalline silicon. The desired purity is the same in both (99.99999%). Molten silicon is poured into molds and allowed to cool. The efficiency and costs of solar cells manufactured from materials produced with this technology are relatively low [11]. In addition, solar cells with an efficiency of around 14% can be obtained from polycrystalline silicon with a particle size of a few millimeters [6].

Monocrystalline or polycrystalline silicon solar cells have been notable for their efficiency and stability since the 1950s. With the technologies developed in recent years, photovoltaic conversion efficiencies are gradually increasing. Production processes are cheaper than monocrystals. Efficiencies of around 18% in laboratory conditions and 14% in commercial modules are obtained from polycrystalline silicon solar cells [1].

It is possible to list the advantages of solar panels today as follows:

- Fuel resources are unlimited;
- They do not produce emissions and radioactive waste. Therefore, it has no effect on global warming;
- Operating costs are low;
- It has no moving parts;
- They can work at ambient temperatures. They do not seek high temperatures;
- They are quite safe. There are panels produced with a guarantee of up to 30 years;

- Annual energy data can be calculated in advance.

2. EXPERIMENTAL DETAILS

Solar cell is defined as the smallest unit that converts sunlight directly into energy in PV systems. The module consists of solar cells connected in series or parallel. By connecting these modules in series and parallel, it is possible to reach PV panels at the desired voltage, power and current values. PV panel converts solar energy into electrical energy with 6%-20% efficiency, depending on the semiconductor material in the panel structure [2]. The most important factor in the operation of PV solar cells is the sunlight falling on the cells. Intensity of sunlight, angle of light falling on the cells; these directly affect the cells' electricity production. Therefore, the latitude and the time of year directly affect the operation of the solar PV system.

Solar energy coming to that region can be estimated according to the geographical location.

Low efficiency of PV panels is related to factors such as temperature, dustiness, panel tilt angle, solar radiation intensity, shading and other losses [7]. The most important parameters affecting panel efficiency are temperature and solar radiation intensity.

Changes in solar radiation intensity and temperature during the day significantly affect panel efficiency, and therefore it is important to know the effects of these parameters on panel efficiency.

However, companies producing PV panels comply with the Standard Test Conditions (STC) of 25 °C cell temperature, 1000 W/m² solar radiation intensity and A.M.

They perform tests under 1.5 air mass ratio conditions and calculate the electrical values of the panel, and the results of these tests are stated in the panel catalogue. The electrical values of the solar panel during changes other than STC are unknown. It is also necessary to know the electrical values of the PV panel under changing atmospheric conditions. Especially under changing atmospheric conditions, calculations made in the design of off-grid and grid-connected systems give more accurate results [8].

The efficiency of solar cells varies depending on the radiation and temperature falling on them. Voltage and current changes depending on the solar radiation falling on the solar cells are shown in Figure 3. From the figure, it can be seen that the change in current due to radiation is large [5].

Panel temperature directly affects the operation of solar cells. As the temperature of the panel increases, a decrease in the efficiency of the panel is observed. Increased temperature also increases the degradation of solar cells and causes a certain decrease in the lifespan of the panels.

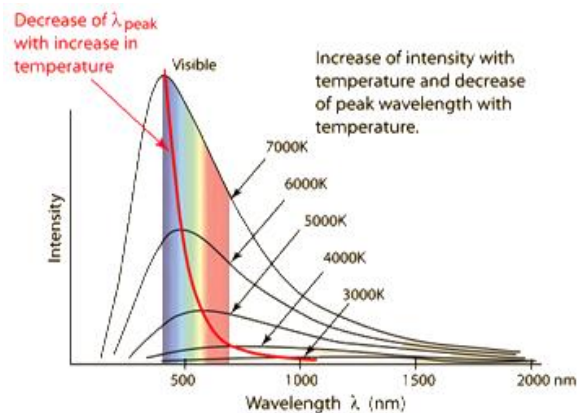


Figure 3. Current and voltage change depending on solar radiation [15]

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The cell temperature (T_c) of a PV panel can be estimated using the NOCT temperature for a desired air temperature (T_a) and irradiance (G) value (kW/m²). μ_p varies between 0.38-0.45 %/K depending on the properties of the panel [4].

$$T_c = T_a + \frac{NOCT - 20}{0.8} G$$

Here G is the radiation value.

The output power of the panel can be calculated by using the temperature value found together with the temperature dependence coefficient μ_p of the panel power. Here P_m , STC is the panel power at standard test conditions (STC) [4].

$$P_m(T_c) = P_{m,STC} [1 - \mu_p (T_c - 25)]$$

$P_m(T_c)$ is the panel power at standard test conditions (STC). NOCT temperature in PV panels varies between 42-52°C. The observed 10°C difference results in a power difference of 3.8% - 4.5%, which is Here, I_{PV} is the current of the solar panel, I_L is the light current, I_s is the diode saturation current, q is the electron charge, V is the voltage of the solar panel, n is the diode factor, k is the Boltzman constant, and T is the cell temperature. Radiation $G=100-1000 \text{ W/m}^2$ and cell temperature T varies between 0 - 75 °C [4].

a significant amount, if the temperature dependence coefficient is accepted as $\mu_p = 0.38 - 0.45 \text{ \%}/K$. Therefore, in panel selection, the NOCT temperature and the temperature dependence coefficient of power should be evaluated together and the ones least affected by temperature should be selected. Thus, the performance decrease will be reduced when the temperature is high [4].

$$I_{PV} = I_L - I_s \left(e^{\frac{qV_{PV}}{nKT}} - 1 \right)$$

In Table 1 in April, the graph of the change in solar radiation value over time and the power at the location where the photovoltaic system is located.

The change graph of production value over time is given. In addition, instantaneous radiation and power production values for 5 different times on April 1 are presented.

Table 1. Some irradiance and power values for April

April						
Hour	Irradiance (W/m2)	Power (kW)	PO	Temperature (°C)	Moisture	Wind Speed (km/h)
09:45	459	23,94	88,3%	12	63%	15
12:00	648,867	34,17	89,1%	15	55%	16
16:00	710	38,86	92,5%	19	37%	22
18:30	114	3,82	56,5%	17	39%	23

Shadowing directly affects the operation of PV panels. The shadow falling on the panel reduces the radiation reaching the PV cells. Due to this decrease in the energy coming to the cells, the cells produce less energy. Another negative effect of shading is the partial shading effect. Even shading of just one cell in the panel can cause the energy produced by the entire panel to decrease significantly. The shaded cell produces very little energy compared to other cells and shows a resistance effect. The electric current produced by other cells begins to pass over the shaded cell and a sudden temperature change is observed in this cell. Shadowing on PV panels does not only cause a decrease in the electricity produced. It also causes

the lifespan of PV panels to decrease due to overheating on the cells due to the effect of shading. The places where PV panels are planned to be installed should be areas that do not receive shade. Care should be taken to ensure that the panels are not affected by the shadow of nearby trees and buildings.

Average radiation, temperature, humidity, wind speed and voltage values for certain hours between 3 and 22 April (Table 2) are given in the example of the condition of weather in Azerbaijan. In this table it is seen that the wind speed value was 26 km/h on April 3 and 2 km/h on April 22.

Table 2. Some values for April 3 and 22

Day	April 3	April 22
Time	11:45	11:15

Irradiance (W/m ²)	975	975
Temperature (°C)	13	13
Humidity (%)	34	34
Wind speed (Km/h)	26	2
Voltage (Volts)	646	616
Current (Ampere)	78,5	78,5
Power production (kW)	50,7	48,4
P.O.	87,9%	83,9%

In the same time period, it is seen that the voltage value on April 3 is higher than the voltage value on April 22, and also, depending on the radiation values, the current value on April 3 is approximately the same as the current value on April 22. When the power production values are compared, the current value is approximately equal and the voltage value is higher. It is thought that this decrease in voltage is due to the increase in panel temperature as the wind speed decreases.

Accumulation of dust particles reduces the performance of solar cells and causes significant losses in power due to solar radiation scattering effects on the surface of the solar panel. It can cause a significant deterioration in the solar conversion efficiency of PV panels. Previous studies on this subject have shown that dust accumulation; It has been shown that the inclination angle of the solar collector is closely related to dust exposure time, site climatic conditions, wind movement and dust properties. However, few studies have also evaluated the impact of PV panel properties, cell types, and surface materials on dust accumulation and efficiency degradation.

1. CONCLUSION

In this study, electricity was produced with PV panels and factors affecting electricity production were examined. Electricity production of PV panels is negatively affected by temperature increase.

In some cases, electrical efficiency decreased despite the decrease in temperature, or electrical efficiency increased as the temperature increased. It has been determined that even a small shadow on the panel series can cause this situation. However, electrical efficiency is also greatly affected by dust.

increased with increasing wind speed, but electricity production decreased due to converter loss, dust and shading of the panel.

The geographical structure of the place where PV panels will be installed is very important. Even a small amount of shading negatively affects the operation of PV panels. Therefore, the place where the PV system will be installed should not be shaded in any way. It has been observed that a PV system installed in a windy place will produce more electricity. However, considering that excessive wind may cause damage to PV panels, it must be installed in a suitable location.

Due to the negative effects of dust, ensuring that the air in the place where PV panels will be installed is clean and less affected by dust will also increase electricity production. It is anticipated that establishing a system to clean dusty panels will also be very useful. It has been determined that converter losses also affect electricity production. For this reason, it is predicted that installing a PV system with a power close to the power of the converter will reduce electricity losses.

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were measured.

The effects of panel temperature, wind speed, surface dust, shading and converter losses were determined. As a result, it was seen that electricity production

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